Thesis Reframing & Defense Action Plan

Strategic Orientation: Computing Education Focus (ACM SIGCSE / ITiCSE Target) | **Defense Deadline:** Nov 19

CORE STRATEGIC SHIFT: THE NEW ANGLE

Proposed Framing Title:

"From Assignment Documents to Reproducible Development Environments: An AI-Assisted Approach for Computer Science Education"

Core Narrative: Move from pure Nix generation to solving a recognized CS education friction point: students wasting time on environment setup rather than learning, and instructors struggling with cross-platform support. Nix is now the implementation mechanism; the assignment-to-environment pipeline is the primary contribution.

ACTIONABLE DEFENSE PREPARATION CHECKLIST

1. NARRATIVE & THESIS REFRAMING

- [ ] **Introduction & Motivation:** Shift framing to student onboarding friction and instructor overhead across heterogeneous hardware.
- [ ] **Pipeline Architecture:** Present the flow: Assignment Doc → AI Extraction → Nix Flake → Validation → Instructor Review → Activation.
- [ ] **Reframe RQs:** Reorganize around RQ1 (AI generation from assignments), RQ2 (Platform reproducibility), and RQ3 (Onboarding benefits & limitations).

2. ASSIGNMENT DOCUMENT EXPERIMENTS (3–5 REAL CS ASSIGNMENTS)

- [ ] Select 3–5 diverse raw CS assignment docs (e.g., Python/Data, Web/Full-stack, Systems/C++).
- [ ] Feed raw documents directly to the agent (no pre-cleaned or structured prompt translation).
- [ ] Evaluate extraction accuracy (languages, tools, dependencies, and pinned versions).
- [ ] Validate generated Nix Flakes and test on clean execution environments.
- [ ] Record failure modes, missing packages, required manual tweaks, and synthesis observations.

3. DATA AUDIT & TECHNICAL REPORTING CLEAN-UP

- [ ] **Student B Data:** Cross-check identical setup times reported in Jupyter vs. Full Stack tables.
- [ ] **Student A Preferences:** Resolve contradictory statements between text sections.
- [ ] **Artifact Sampling:** Document explicit selection criteria for the 41 clean-replay artifacts.
- [ ] **Metric Precision:** Clearly separate environment *activation* success from functional *task execution*.
- [ ] **Claim Calibration:** Tone down MCP claims (reflect mixed synthesis accuracy) and qualify Nix setup overhead.

4. DEFENSE DECK & PRESENTATION DELIVERY

- [ ] **Hook Slide:** Open with an authentic, complex CS assignment doc and transition to the one-command activated environment.
- [ ] **Evidence Mapping:** Present existing benchmark + macOS/Linux replays + new doc experiments as evidence for RQs 1–3.

POST-DEFENSE ROADMAP (SIGCSE / ITICSE TARGET)

- [ ] Expand real-world assignment trials across additional courses and run a broader empirical classroom onboarding study.